# Mikhail Seleznyov

simplymike8@gmail.com Mikhail Seleznyov Dont-Care-Didnt-Ask

AI Safety Researcher — adversarial and non-adversarial LLM robustness, misalignment, uncertainty quantification.

## Work & Internships

### AIRI Institute — Oct 2024 – Present time

*Research Scientist*

- Supervised a study asking whether emergent misalignment – where narrow adaptation on a limited domain broadly corrupts model behavior – can arise through in-context learning rather than only finetuning. Across 4 frontier model families, misaligned responses emerged from as few as 2 in-context examples, with larger models generally more susceptible. The paper was accepted to ACL 2026.
- Led the design of an LLM-powered evolutionary search framework that automatically discovers unsupervised uncertainty quantification (hallucination detection) methods, represented as Python programs. Our evolved methods outperformed strong hand-designed baselines by up to 6.7% relative ROC-AUC across 9 datasets on atomic claim verification, despite only being optimzed for a single dataset. The paper was accepted to ACL SRW 2026.
- Compared prompt sensitivity mitigation methods for language models. I came up with the initial idea, lead the project, conducted experiments and wrote a substantial part of the paper. We found out that 1) test-time probability calibration increases robustness but is brittle to class imbalance; 2) finetuning with augmentations is surprisingly ineffective for increasing robustness; 3) in ensembling, majority voting is more stable than averaging predicted probabilities. The paper was accepted to EMNLP Findings 2025. Link to code.
- Developed efficient neural metrics for machine translation. I conducted experiments with quantization and pruning methods, and co-designed the training procedure for distillation. We reduced memory footprint up to x3 times without any degradation, and up to x32 with a moderate quality drop. Corresponding paper was accepted to EMNLP Main 2024. Link to code.

### Center for Human-Compatible AI — July 2024 – November 2024

*Research Intern*

- CHAI is a multi-institute research organization based out of UC Berkeley that focuses on foundational research for AI technical safety.

  I worked with Scott Emmons (Google DeepMind) on the topic of inconspicuous adversarial attacks. Resulting paper: Obfuscated Activations Bypass LLM Latent-Space Defenses. Accepted to ICLR 2026.

### ML Alignment & Theory Scholars 5.0 — January 2024 – May 2024

*Mentee*

- The ML Alignment & Theory Scholars (MATS) Program is an highly selective (< 9% average per-mentor acceptance rate) independent research and educational seminar program held in Berkeley, California which focuses on AI Safety problems.

  I worked with David Lindner (Google DeepMind) on benchmarking vision-language models as reward models for RL. Resulting preprint: ViSTa Dataset: Do vision-language models understand sequential tasks?

### Yandex — Oct 2021 – Oct 2024

*Research Scientist*

- Graph-based and long-term time series forecasting. Implemented graph-agnostic baselines and designed experiments on the importance of long context windows and graph aggregations; gave a talk on SOTA methods at a Yandex Laboratory seminar (slides). Co-authored a NeurIPS 2025 NPGML Workshop paper.
- Self-supervised pretraining for computer vision: compared and adapted Masked Autoencoder, SimMIM, and data2vec for distributed-setting scaling on LAION-400M (report), and compared architectures/pretraining objectives for instance segmentation and object detection.

## Education

**Skoltech** **Sep 2024 – Present time**
*PhD Data Science*

**Skoltech** **Sep 2022 – June 2024**
*MSc Data Science*
***GPA: 5.0/5.0**, graduated with Honors*

**Yandex School of Data Analysis** **Sep 2020 – June 2022**
*MSs-level additional program, Machine Learning developer track*
***GPA: 5.0/5.0**, graduated with Honors*

**Moscow State University** **Sep 2018 – June 2022**
*BSc Applied Mathematics and Computer Science*
***GPA: 4.95/5.0**, graduated with Honors*

## Selected publications (full list at Google Scholar)

**ACL 2026**
- Emergent Misalignment via In-Context Learning: Narrow in-context examples can produce broadly misaligned LLMs.

**ACL SRW 2026**
- Evolutionary Search for Automated Design of Uncertainty Quantification Methods.

**ICLR 2026**
- Obfuscated Activations Bypass LLM Latent-Space Defenses.

**EMNLP 2025 Findings**
- When Punctuation Matters: A Large-Scale Comparison of Prompt Robustness Methods for LLMs.

**EMNLP Main 2024**
- xCOMET-lite: Bridging the Gap Between Efficiency and Quality in Learned MT Evaluation Metrics.

**Preprint**
- Breaking the Chain: A Causal Analysis of LLM Faithfulness to Intermediate Structures.

**PACLIC, 2023**
- A computational study of matrix decomposition methods for compression of pre-trained Transformers.

## Technical Skills

**Languages:**
- Python ($\sim$ 4000+ hours of coding), C/C++ ($\sim$ 400 hours), R ($\sim$ 60 hours).

**Developer Tools:**
- VS Code, Cursor, Git, Linux shell, Docker.

**Techologies/Frameworks:**
- GitHub, PyTorch, HuggingFace, vLLM, numpy/pandas/sklearn/joblib/scipy/seaborn, hydra, dvc.

## Achievements

**"Major League" Applied Mathematics and Computer Science Olympiad** **Spring 2022**
- Medalist — shared 1 place out of 293 finalists in a national olympiad.

**International Data Analysis Olympiad** **March 2022 – April 2022**
- Finalist — member of the team "Mylene Farmer".